

Modelling with MATLAB Assignment 5

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1 Deterministic SIR model of epidemic.

Consider a version of the SIR model of an epidemic, given by

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{IS}{N}, \\ \frac{dI}{dt} &= \beta \frac{IS}{N} - \gamma I - \mu I, \\ \frac{dR}{dt} &= \gamma I, \\ N &= S + I + R,\end{aligned}\tag{1}$$

where $S(t)$, $I(t)$ and $R(t)$ are the numbers of susceptible, infected and recovered individuals; β is a coefficient characterizing the transmission of infection from an infected individual to a susceptible one; γ is the recovery rate and μ is the mortality rate due to infection. Let the parameter values be as follows:

$$\beta = 2, \quad \gamma = 0.1, \quad \mu = 0.01.\tag{2}$$

1.1 (a)

Solve the system with these parameters using MATLAB's **ode45** with the initial conditions $S(0) = 100$, $I(0) = 1$ and $R(0) = 0$ and plot the results.

I created the function “sirfun.m” from Equation 1. This function takes a time variable t and a vector “ x ” and has an output of a 1×3 array.

I created the script “sirmodelode1.m” to set the global parameters needed for the fixed values β, γ and μ then set the span of t to be between 0 and 100 and the initial value $x_0 = [100, 1, 0]$. I then used the

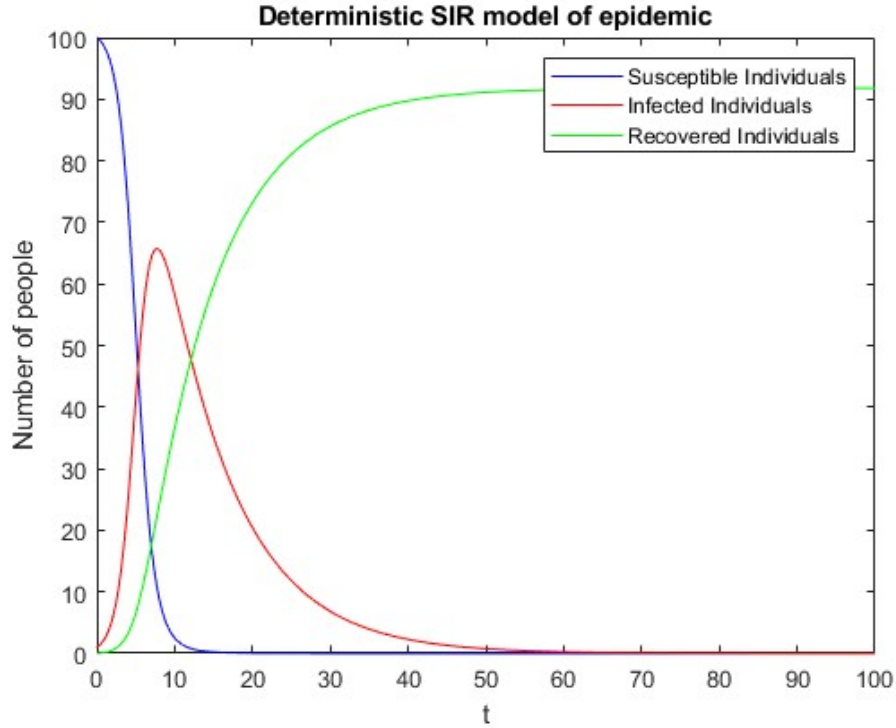


Figure 1: Question 1a - Deterministic SIR model of epidemic.

ode45 solver and plotted all values of x against t as seen in Figure 1. This graph shows that the susceptible individuals (in blue) quickly depletes, the infected individuals peaks at $t = 7$ with about 65 people (in red) and the number of recovered individuals rises to a value of approximately 90, meaning 10 people died.

1.2 (b)

Compute and plot the total number of deaths due to epidemic for a range of values of β . Keep all the other parameters and initial conditions fixed.

The script file “sirmodelode2.m” sets global parameters γ and μ I also create an array spanning values for which I would like β to take. I chose $1 \leq \beta \leq 25$. Then for these 25 values, I ran them through the ode45 solver and found the number of deaths at the end time for each value. I then plotted these death values as seen in Figure 2. The number of deaths are not affected by changing the value of β .

I also had the script “sirmodelode2.m” choose 4 random integers and plotted the graphs of these SIR models. These can be seen in Figure 3. I looked at when these graphs were between $0 \leq t \leq 10$ and the value of β changes how quickly people become infected but not the number of deaths. The number of deaths seen is around 9.1.

2 Stochastic SIR model of epidemic.

2.1 A.

Create MATLAB code which simulates an epidemic as a sum of three independent Poisson processes:

Poisson process	Rate
Infection of susceptible individual	$\beta SI/N$
Death of infected individual	μI
Recovery of infected individual	γI

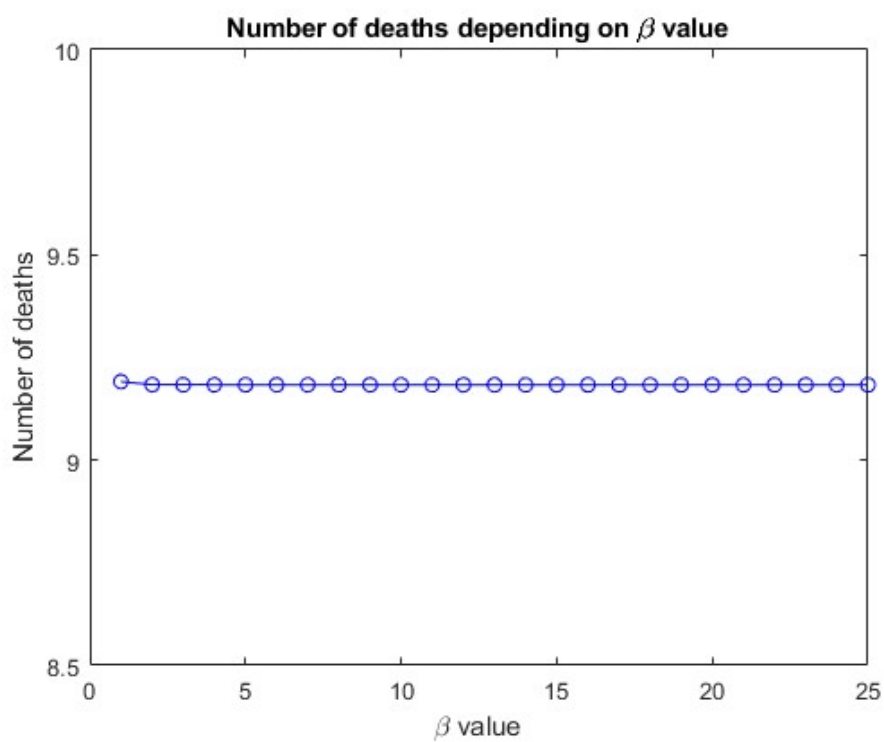


Figure 2: Question 1b - Number of deaths depending on β values.

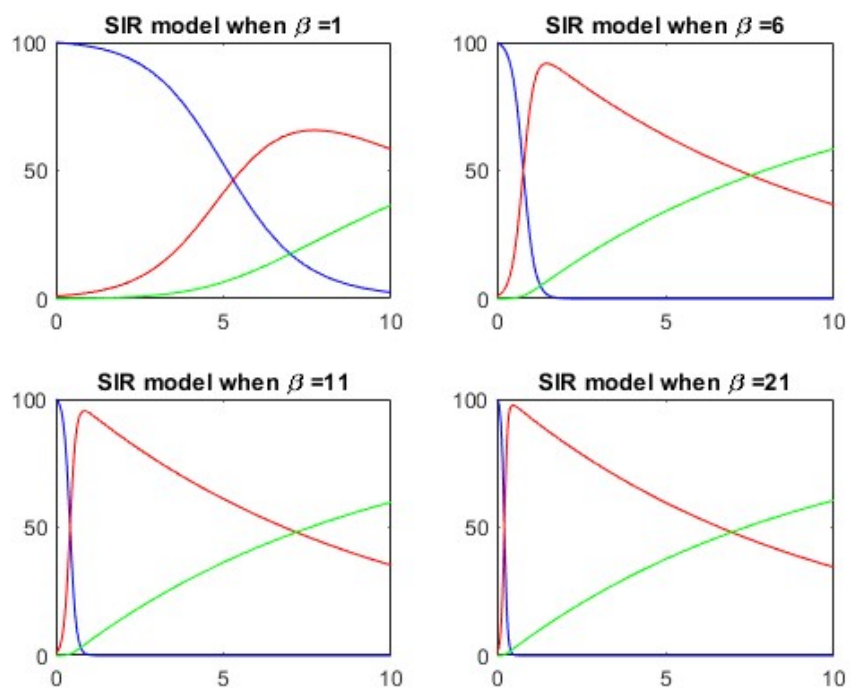


Figure 3: Question 1b - SIR plots with varying β values.

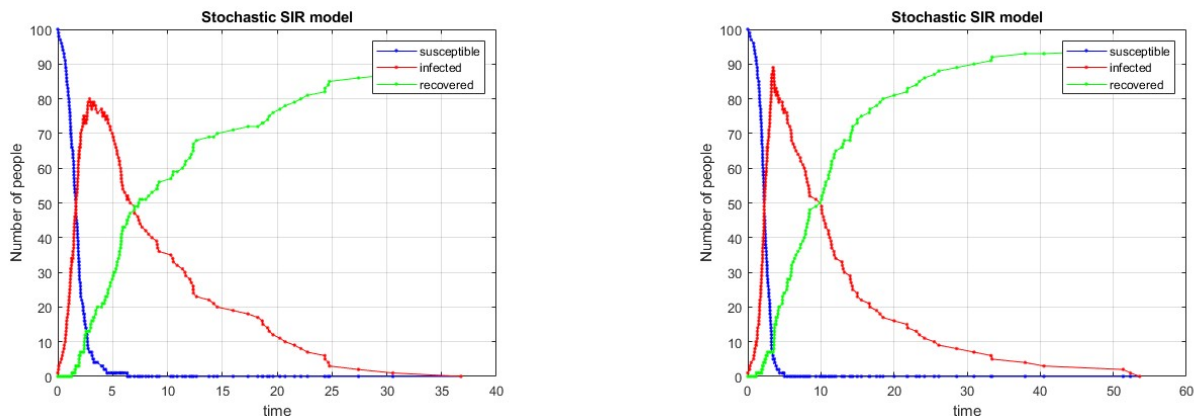


Figure 4: Stochastic SIR model graphs.

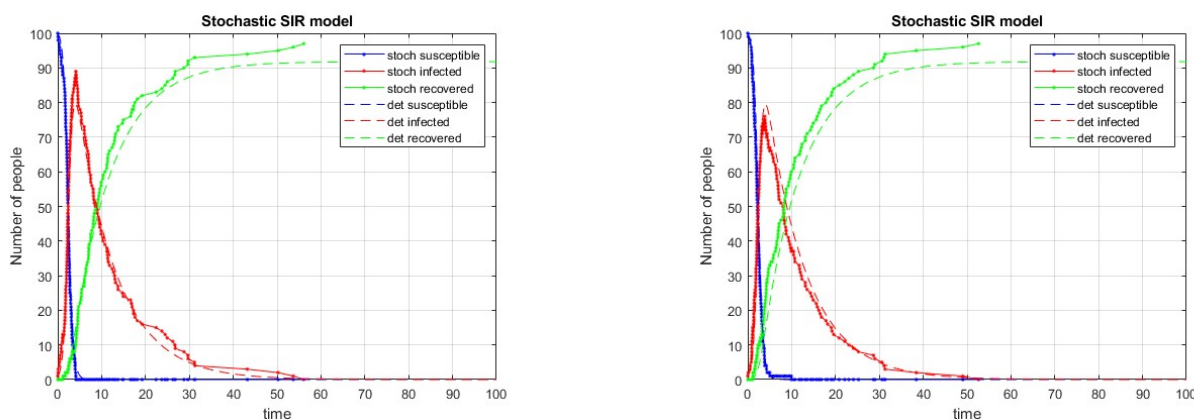


Figure 5: Stochastic SIR model graphs with comparison to Deterministic SIR Model.

Simulate this process with parameter values given by Eq. 2 and initial conditions $S(0) = 100$, $I(0) = 1$ and $R(0) = 0$. Provide graphical output showing two independent examples of the process and compare it with the deterministic model of question 1.

I modeled the Stochastic SIR model in the function “SIR_stoch_poisson.m” and then created the additional function “SIR_stoch_poisson_run.m” to script the inputs for the model. My script creates 4 graphs to show the Stochastic SIR model, it creates 2 which just show the model by itself - Figure 4 - and 2 which show both the stochastic model and the deterministic model - Figure 5.

2.2 B.

For a suitably large sample size,

2.2.1 (i)

calculate the mean duration of the epidemic (i.e. the time needed for the number of infected individuals to drop to zero) and plot the histogram;

I chose a sample size of 500. The mean duration of the epidemic for my run was 46.059. I created the histogram to show this data - Figure 6. On 33 occasions the epidemic died out very quickly but on most other occasions the epidemic lasted longer than a time of 35.

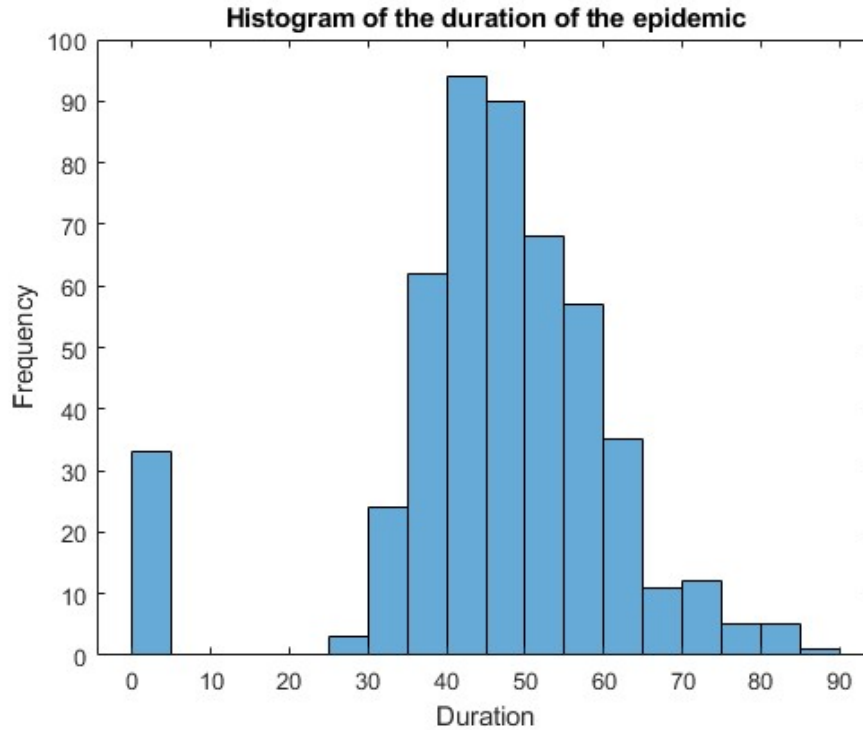


Figure 6: Question 2b i - Histogram of the mean duration of the epidemic.

2.2.2 (ii)

calculate the mean of the total number of deaths due to the epidemic and compare it with the prediction of the deterministic model of question 1.

The mean total number of deaths due to the epidemic was 8.7280. The histogram in Figure 7 shows the frequency of the number of deaths. The histogram shows that on 28 occasions no one died.

2.2.3 (iii)

calculate the mean time needed for the epidemic to reach its peak (i.e. the maximum number of infected individuals);

The mean time needed for the epidemic to reach its peak is 3.8947. The histogram in Figure 8 shows the duration of time needed for the epidemic to reach its peak.

2.2.4 (iv)

approximate the probability that the number of infected individuals at the peak will be more than 10;

The probability that the number of infected individuals at the peak will be more than 10 is 93.4%. I found this by taking the index for when the epidemic reached its peak and working out whether this was greater than 10. I then counted the number of times this occurred before dividing by the number of trials and timesing by 100.

2.2.5 (v)

if there is a vaccine which 100% effective, what level of vaccination (a percentage of vaccinated population) is needed to reduce the probability computed in part (iv) by a factor of 2?

I created a separate script to answer this question, "SIR_stoch_run_vaccination.m". This script ran 500 runs and figured out the probability that the number of infected individuals is less than 10. Then it took

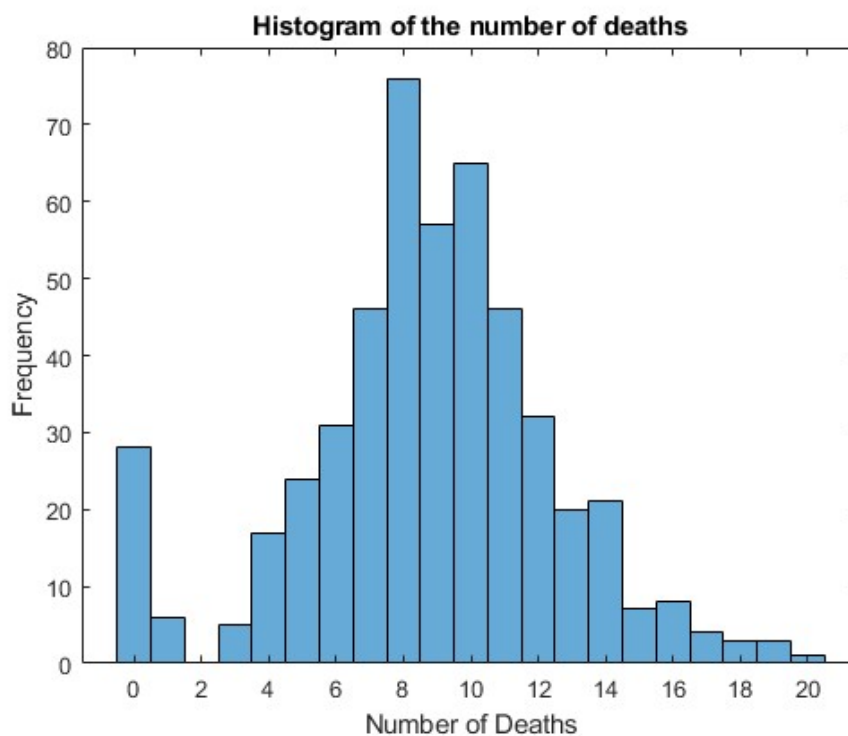


Figure 7: Question 2b ii - Histogram of the total number of deaths due to the epidemic.

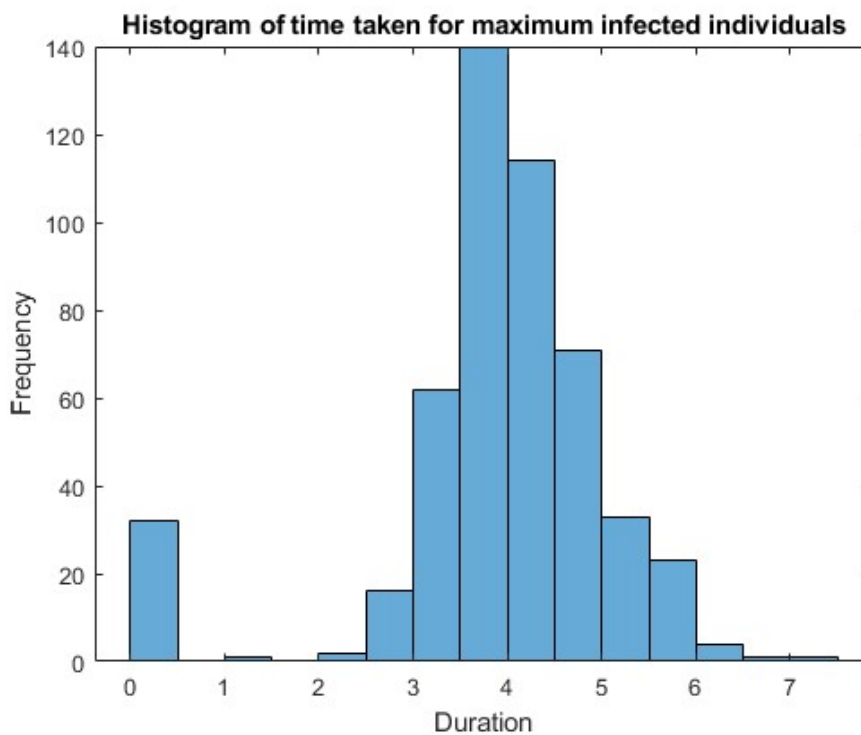


Figure 8: Question 2b iii - Histogram of the mean time needed for the epidemic to reach its peak.

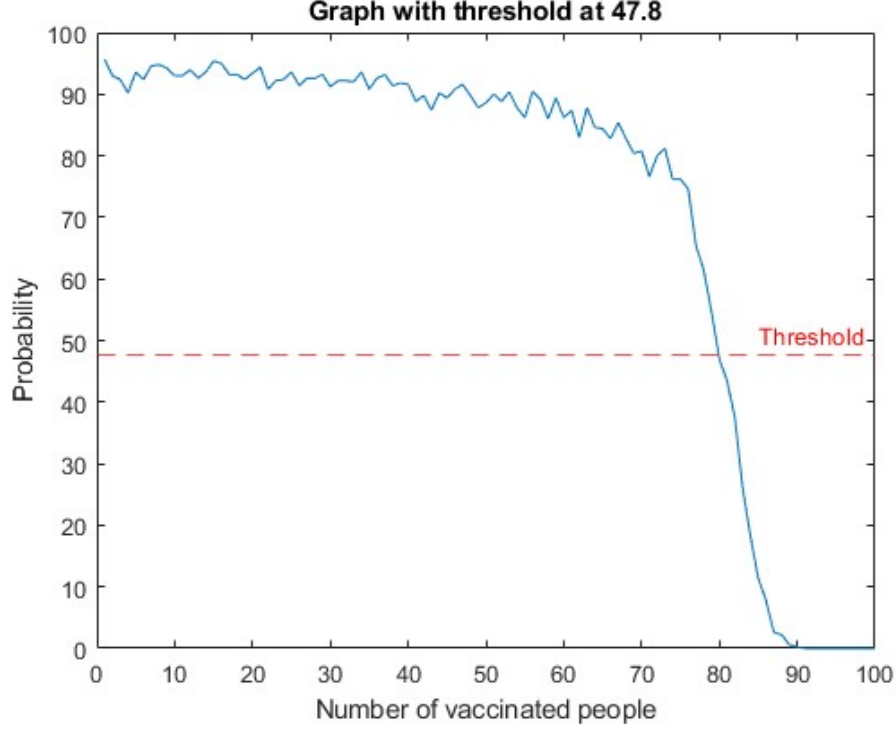


Figure 9: Question 2b v - Graph showing the number of vaccinated people against the probability that the number of infected individuals will peak at more than 10.

one person from the susceptibles and added them to the recovered as the new initial conditions and ran the script again. I did this 100 times.

I took the value of the 1st run and divided by 2 to figure out what percentage I was aiming for. I found out where this position was then plotted all the individual percentages on a graph as seen in Figure 9. I added a line to this graph to show the threshold for when the percentage is less than a factor of 2. For me the threshold was 47.8% and this started at 80 vaccinated people.

3 M-Level - SIS model of infection with logistic growth.

Assume that (i) without infection, the population growth is described by the logistic equation and there is natural mortality, and (ii) the recovered individuals have no immunity (i.e. susceptible to infection). These assumptions lead to the following dynamical system:

$$\begin{aligned}\frac{dS}{dt} &= rS \left(1 - \frac{S}{K}\right) - \beta \frac{IS}{S+I} - \mu S + \gamma I, \\ \frac{dI}{dt} &= \beta \frac{IS}{S+I} - (\mu + \gamma + \delta)I,\end{aligned}\tag{3}$$

where $S(t)$ and $I(t)$ are the numbers of susceptible and infected individuals, β is a coefficient characterizing the transmission of infection from an infected individual to a susceptible one, r is the growth rate of population, K is the carrying capacity; γ is the recovery rate, μ is the natural death rate, and δ is the death rate due to infection. The parameter values are fixed:

$$r = 0.2, \quad K = 100, \quad \beta = 0.25, \quad \gamma = 0.01, \quad \mu = 0.01, \quad \delta = 0.02.\tag{4}$$

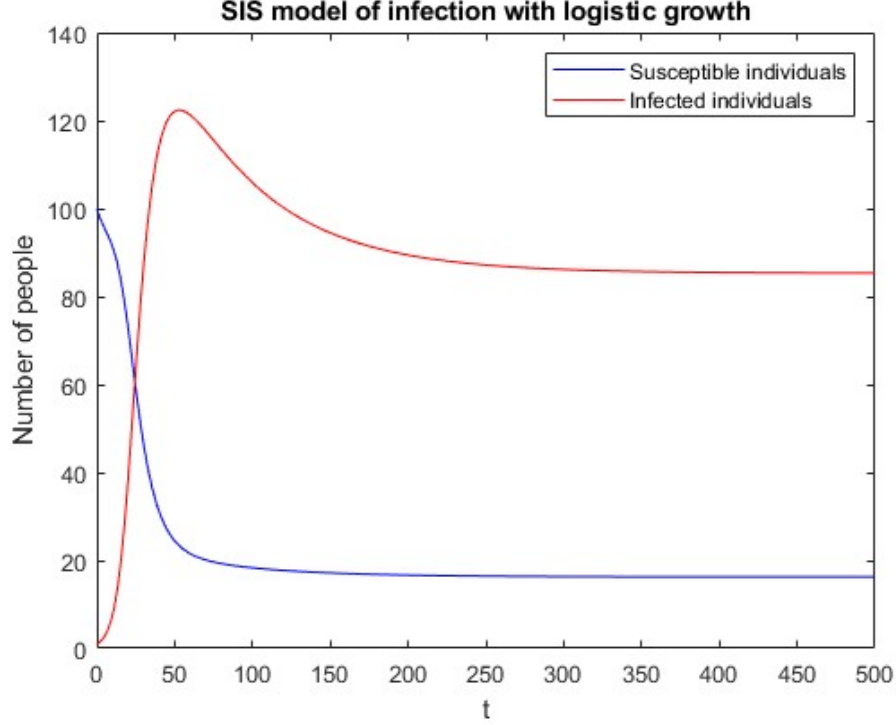


Figure 10: Question 3a - SIS model of infection.

3.1 A.

Solve this system with initial conditions $S(0) = 100$, $I(0) = 1$ using MATLAB and show that the system has an endemic equilibrium where both S and I are nonzero.

I put the system in equation 3 into the function “sisfunA.m”. This function takes initial values t and an array x and returns a 2×1 array. To solve the system I first declared global variables r, K, β, γ, μ and δ . I then set the span of t to be between 0 and 500 and the initial condition $x_0 = [100; 1]$. I then used the ode45 solver and then plot the results which can be seen in Figure 10.

The figure shows the infected individuals peak at around $t = 50$ with around 120 individuals and then lowers. Both the susceptible and infected individuals reach an endemic equilibrium with the number of infected individuals levelling to about 85 people and the number of susceptible people levelling to about 16 people.

3.2 B.

Suppose that, for each individual, there is a delay between contracting the infection and being able to transmit it to others. Show how such a delay could be incorporated into this dynamical system, stating clearly any assumptions you make. For your new delay-differential equations with the parameter values given by (4), use MATLAB to investigate whether the addition of a delay can cause instability in the system, resulting in periodic solutions.

In order to adapt the system in Equation 3 to include a delay you need to use dde23 solver. I adapted the equations to be as follows:

$$\begin{aligned} \frac{dS}{dt} &= rS(t) \left(1 - \frac{S(t)}{K(t)} \right) - \beta \frac{I(t)S(t)}{S(t) + I(t)} - \mu S(t) + \gamma I(t) + \gamma I(t - \tau), \\ \frac{dI}{dt} &= \beta \frac{I(t - \tau)S(t - \tau)}{S(t - \tau) + I(t - \tau)} - (\mu + \gamma + \delta)I(t - \tau) - (\mu + \gamma + \delta)I(t). \end{aligned} \quad (5)$$

This system assumes that people who have the infection can die and can recover before becoming infectious.

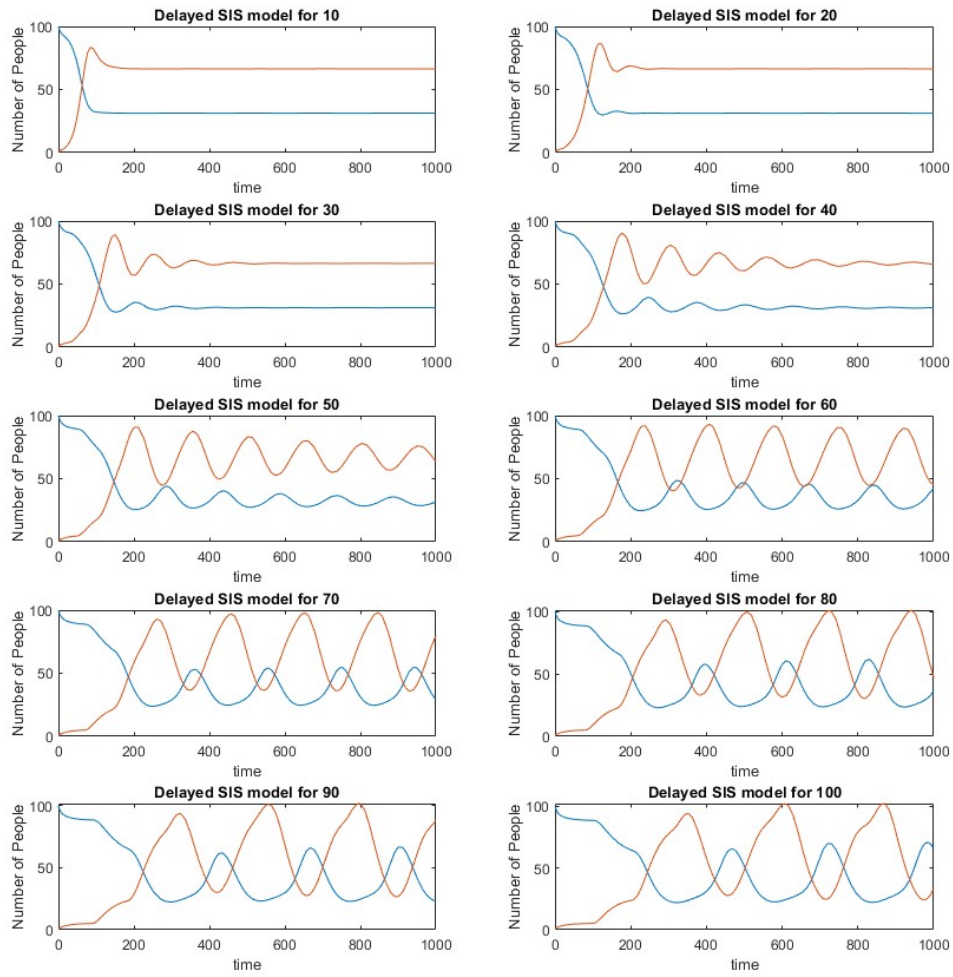


Figure 11: Question 3b - SIS delayed Model: Blue - Susceptible, Red - Infected.

I modelled the system with delays from 10 to 100 in increments of 10. As seen in Figure 11, as you increase the delay, the instability of the system increases. The resulting solutions are periodic.